Multimodal Reasoning for Shrimp Disease Diagnosis in Edge IoT Systems

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Abstract. Shrimp disease diagnosis in real aquaculture environments is intrinsically uncertain: visual evidence is degraded by turbidity and occlusion, while farmer-reported symptoms are brief, informal, and semantically overlapping. Existing AI systems largely assume unimodal sufficiency, overlooking the fact that practical diagnosis is a multimodal disambiguation problem under heterogeneous noise. We formalize shrimp disease detection as uncertainty-aware multimodal inference, where effective fusion should reduce predictive entropy relative to unimodal models. To this end, we propose a cross-modal attention framework built upon transferable vision and language foundation models, specialized via parameter-efficient fine-tuning. The model adaptively reweights modalities according to reliability and is deployed within an Edge–Fog–Cloud IoT architecture using structured pruning and quantization for low-latency inference. Experiments on 2,650 field images and 10,530 farmer reports demonstrate 98.2% macro F1, significantly outperforming unimodal and late-fusion baselines. Entropy–attention analysis confirms systematic uncertainty reduction, validating scalable multimodal reasoning for cyber-physical environmental data mining.

Keywords: Efficient foundation models · Aquaculture disease diagnosis · Large-scale environmental data mining.

1 Introduction

Global aquaculture now exceeds capture fisheries in total production, with shrimp farming playing a major economic role [1]. However, diseases such as White Spot Syndrome Virus (WSSV) and Acute Hepatopancreatic Necrosis Disease (AHPND) continue to cause severe losses, especially in intensive Southeast Asian systems [2]. Because mortality can escalate within days, early diagnosis is critical. In practice, diagnosis occurs under highly imperfect conditions. Farmers observe shrimp in turbid water and low lighting, where lesions are partially occluded and camera images are unclear. At the same time, they report short textual symptoms such as “tôm ăn kém,” “bơi lờ đờ,” or “phân trắng nổi.” These descriptions are informal and overlap across multiple diseases. For instance, reduced feeding may indicate viral infection, bacterial infection, or environmental stress.

Shrimp disease diagnosis is therefore **inherently multimodal and uncertain**. It is not purely a vision problem, since visual evidence is noisy and incomplete. It is not purely a language problem, since farmer reports are imprecise. Instead, it constitutes a **multimodal disambiguation problem**, requiring joint reasoning over degraded visual cues and ambiguous human language within a cyber-physical aquaculture system.

Most existing AI systems in aquaculture operate under unimodal assumptions. CNN-based image classifiers achieve promising accuracy on curated datasets [3], yet their robustness degrades under low visibility and field conditions. Text-based disease classification using TF-IDF or deep neural models shows strong performance on standardized symptom corpora [4], but struggles when farmer language is short, vague, or inconsistent. Besides, IoT monitoring platforms primarily rely on environmental threshold rules for parameters such as pH, dissolved oxygen, and salinity [5]. While effective for anomaly detection, these systems lack semantic reasoning about disease categories and do not integrate visual or linguistic evidence.

Although multimodal learning has advanced rapidly with foundation models [6], existing systems rarely address uncertainty resolution between noisy modalities, and even fewer studies consider **efficient deployment of vision–language foundation models on resource-constrained edge devices** in large-scale IoT environments. The integration of multimodal reasoning with edge-efficient inference remains underexplored in aquaculture AI.

This paper makes three key contributions:

1. **Problem Formalization.** We formalize shrimp disease diagnosis as a multimodal ambiguity resolution problem, where noisy visual observations and imprecise farmer language must be jointly interpreted under environmental uncertainty.
2. **Efficient Multimodal Reasoning Framework.** We propose a cross-modal attention framework integrating vision and language foundation models to resolve diagnostic ambiguity beyond simple late fusion.
3. **Edge-Optimized and Real-World Validation.** We deploy the system within an Edge–Fog–Cloud IoT architecture using pruning and quantization for efficient inference, and validate it on large-scale field data streams comprising real pond images and farmer reports.

By reframing shrimp disease detection as multimodal reasoning under uncertainty and demonstrating efficient foundation model deployment in cyber-physical systems, this work contributes a practical blueprint for scalable environmental intelligence.

2 Background and Related works

Shrimp disease diagnosis has traditionally relied on laboratory-based assays such as PCR, which offer high accuracy but are unsuitable for real-time, large-scale pond monitoring. Recent AI approaches primarily formulate disease detection as a supervised image classification problem. CNN-based models with transfer learning have achieved competitive performance on curated shrimp image datasets [3]. Edge-optimized architectures further demonstrate that lightweight CNN backbones can operate in real time on Jetson-class devices [7]. In parallel, text-based disease identification has been explored using TF-IDF and deep neural models to classify symptom descriptions [4]. Meanwhile, IoT-driven frameworks focus on anomaly detection in water-quality streams using autoencoders and forecasting models [8]. However, these lines of work share a fundamental limitation: they treat disease inference as a **unimodal learning problem**. Vision-only models assume visual sufficiency; text-only models assume linguistic clarity; anomaly detection systems ignore semantic disease categories. None explicitly model the **diagnostic ambiguity arising from cross-modal inconsistencies in real farming conditions**. From a data mining perspective, shrimp disease detection remains framed as isolated classification or anomaly detection tasks rather than a structured multimodal reasoning problem under uncertainty.

Multimodal learning has evolved from simple late fusion schemes to cross-modal attention mechanisms and large-scale vision–language foundation models. Late fusion concatenates independently learned features but does not capture inter-modal dependencies. Cross-modal attention improves alignment by allowing one modality to reweight another, significantly enhancing semantic interaction in vision–language tasks. Foundation models further enable cross-domain transfer through pretrained representations. However, their success has largely been demonstrated on web-scale benchmarks with abundant supervision and high computational budgets. In contrast, cyber-physical IoT environments impose strict constraints on latency, memory, and energy consumption. In aquaculture, AIoT surveys emphasize the potential of integrating sensing and AI but highlight the absence of scalable multimodal intelligence in operational farms [9]. Crucially, existing multimodal systems rarely formalize the problem as **uncertainty resolution between noisy modalities**, nor do they evaluate performance under field-level edge deployment constraints.

Efficient foundation models aim to retain transferability while reducing computational cost through pruning, quantization, and mixed-precision execution. Edge deployments in aquaculture demonstrate feasibility for unimodal CNN inference [7] and lightweight anomaly detection over streaming sensor data [8]. Yet, integrating **efficient multimodal foundation** models within real IoT pipelines remains largely unexplored. Existing systems either (1) optimize efficiency without multimodal reasoning, or (2) adopt multimodal architectures without considering edge constraints.

From the standpoint of large-scale data mining, there is a clear gap:

1. No formalization of shrimp disease detection as a **multimodal ambiguity resolution problem.**
2. No unified framework combining **vision–language foundation models, edge-efficient optimization,** and **continuous IoT data streams**.
3. No empirical validation of multimodal foundation models in operational aquaculture environments.

Addressing this gap requires not merely combining modalities, but rethinking shrimp disease diagnosis as **efficient multimodal reasoning under real-world uncertainty**.

3 Methodology

3.1 Problem Formulation

Multimodal Diagnostic Setting We model shrimp disease diagnosis as multimodal inference under uncertainty. Each observation consists of:

- an image $I \in \mathbb{R}^{H \times W \times 3}$,
- a textual report $T = \{w_1, \dots, w_n\}$,
- optionally sensor context $S \in \mathbb{R}^d$,

with the objective of predicting a disease label $y \in \mathcal{Y}$. Unlike standard multimodal benchmarks, both modalities may be independently noisy and semantically incomplete. The core challenge is to infer y when either visual or linguistic evidence is unreliable.

Ambiguity and Uncertainty Modeling Let $p_{\text{img}}(y \mid I)$ and $p_{\text{text}}(y \mid T)$ denote unimodal predictive distributions. We quantify modality-specific uncertainty using Shannon entropy [10]:

$$H_{\text{img}} = -\sum_{y \in \mathcal{Y}} p_{\text{img}}(y \mid I) \log p_{\text{img}}(y \mid I), \tag{1}$$

$$H_{\text{text}} = -\sum_{y \in \mathcal{Y}} p_{\text{text}}(y \mid T) \log p_{\text{text}}(y \mid T). \tag{2}$$

A case is considered ambiguous when either H_{img} or H_{text} is high. From an information-theoretic perspective, effective fusion should maximize the conditional mutual information between modalities and the label:

$$\max_f \; I(Y; I, T \mid S), \tag{3}$$

while reducing predictive uncertainty relative to unimodal inference:

$$H(Y \mid I, T, S) < \min \{H(Y \mid I),\; H(Y \mid T)\}. \tag{4}$$

We therefore seek a fusion function

$$f(I, T, S) \to y \tag{5}$$

that minimizes expected classification loss while explicitly resolving modality-specific ambiguity through cross-modal interaction.

3.2 System Architecture

Figure 1 presents the proposed Edge–Fog–Cloud IoT architecture for multimodal shrimp disease diagnosis. The design follows a scalable cyber-physical deployment paradigm, enabling low-latency inference at the edge while supporting centralized storage and retraining in the cloud.

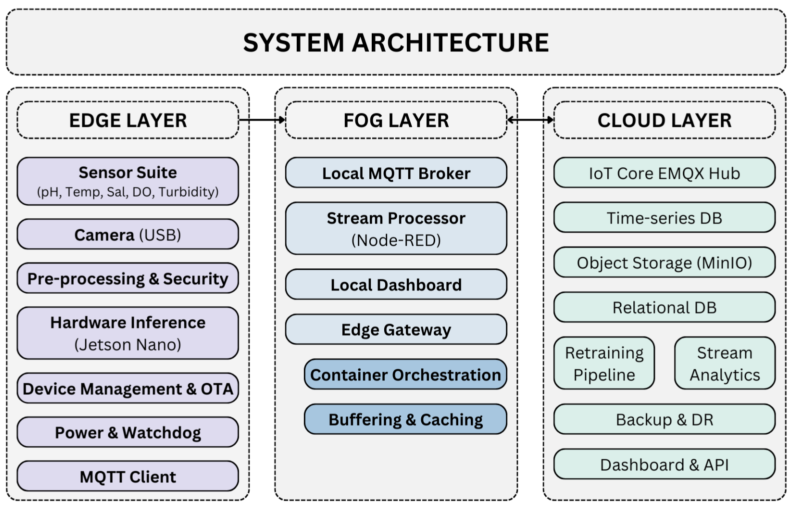

Fig. 1. Edge–Fog–Cloud architecture for multimodal shrimp disease diagnosis.

Edge–Fog–Cloud IoT Framework

The edge node performs real-time sensing and inference. A sensor suite continuously samples environmental parameters (pH, temperature, salinity, dissolved oxygen, turbidity), while a USB camera captures shrimp images. Data are preprocessed locally and forwarded to a Jetson Nano device for hardware-accelerated multimodal inference. Device management, OTA updates, and watchdog mechanisms ensure robustness under field conditions.

The fog layer acts as a coordination and buffering hub. A local MQTT broker handles secure message exchange between edge devices and upstream services. Stream processing and lightweight orchestration manage temporary storage, aggregation, and local dashboard visualization, reducing uplink bandwidth and improving resilience during network disruptions.

The cloud layer provides scalable storage and model lifecycle management. Time-series databases and object storage retain sensor streams and image data. A retraining pipeline supports periodic model updates using accumulated field data, while APIs and dashboards expose monitoring and analytics services. OTA updates propagate optimized models back to edge devices.

Overall, the architecture enables secure, low-latency, and scalable deployment of multimodal foundation models in distributed aquaculture environments.

Data Pipeline The system operates on real-world multimodal streams comprising 2,650 field images and 10,530 farmer text reports. Data are ingested through MQTT channels, processed in streaming mode, and forwarded to the edge inference engine. Predictions are generated in real time and trigger automated alert notifications when disease probability exceeds predefined thresholds. Simultaneously, all multimodal observations are archived in the cloud to support continual retraining and performance monitoring.

3.3 Multimodal Learning Framework

Vision Foundation Model We adopt ResNet-50 [11] pretrained on ImageNet as a transferable vision foundation backbone. Rather than training from scratch, we leverage large-scale pretraining to obtain domain-agnostic visual representations and adapt them to shrimp pathology via lightweight fine-tuning. This parameter-efficient adaptation preserves general visual priors while requiring only a small task-specific update. Importantly, our objective is not to design a larger vision architecture, but to investigate how pretrained foundation representations can be efficiently specialized and deployed in resource-constrained cyber-physical environments. Structured pruning and FP16 quantization are applied post-adaptation to retain transferability while meeting strict edge latency constraints. The encoder outputs a transferable feature embedding $f_{\text{img}} \in \mathbb{R}^{d_v}$.

Language Foundation Model For textual reports, we employ PhoBERT-base [12] as a pretrained language foundation model trained on large-scale Vietnamese corpora. Instead of constructing a task-specific text classifier from scratch, we exploit pretrained contextual representations and perform parameter-efficient fine-tuning to adapt semantic priors to aquaculture terminology. This setup emphasizes transferability: general linguistic knowledge learned from broad corpora is specialized to low-resource, domain-specific symptom descriptions without large-scale in-domain pretraining. The contextual embedding of the [CLS] token serves as a global semantic representation, producing $f_{\text{text}} \in \mathbb{R}^{d_t}$.

Late Fusion Baseline As a baseline, we implement late fusion via feature concatenation, where the joint representation is defined as $f = [f_{\text{img}} \mid\mid f_{\text{text}}]$, followed by a classification head. This approach assumes independent modality contributions and does not explicitly model cross-modal interactions.

Cross-Modal Attention for Ambiguity Resolution To resolve diagnostic ambiguity, we introduce cross-modal attention to model inter-modal dependencies. For feature component i, the attention score is computed as

$$e_i = v^T \tanh\left(W_{\text{img}} f_{\text{img},i} + W_{\text{text}} f_{\text{text},i}\right), \tag{6}$$

followed by softmax normalization to obtain adaptive weights. The fused representation is a weighted combination of modality features. This mechanism dynamically reweights modalities based on their reliability. In high-entropy visual cases, linguistic features receive higher attention, and vice versa, enabling explicit ambiguity reduction beyond static fusion.

Efficient Deployment To ensure scalability, we compress both encoders via pruning and quantization. The optimized multimodal model achieves sub-200 ms inference latency on Jetson Nano devices. Local inference reduces uplink transmission to predictions and compact metadata, resulting in approximately 70% bandwidth reduction.

Foundation Model Perspective From a foundation model standpoint, our contribution lies not in scaling model size, but in demonstrating that pretrained vision and language models can serve as transferable representation engines for domain-specific multimodal reasoning. We show that (1) large-scale pretrained priors can be specialized through lightweight fine-tuning, (2) cross-modal interaction reduces uncertainty without increasing parameter count substantially, and (3) compression techniques preserve predictive performance while enabling edge deployment. This establishes a generalizable template for efficient foundation model adaptation and deployment in cyber-physical IoT systems.

4 Experimental Results

4.1 Experimental Setup

We evaluate the proposed multimodal framework on a real-world shrimp farming dataset collected under operational pond conditions. The image dataset contains 2,650 field images across four classes (BG, Healthy, WSSV, WSSV-BG), captured under turbidity, varying illumination, and non-frontal viewing angles. The text dataset comprises 10,530 farmer-derived symptom reports spanning twelve disease categories. Both datasets are stratified into 80/10/10 train, validation, and test splits to preserve class balance. All models are trained on an NVIDIA A100 (80GB) GPU using mixed precision. Vision models are optimized with Adam and cosine annealing (initial learning rate 10^{-4}, 80 epochs). PhoBERT-base is fine-tuned using Adam (2×10^{-5}) with early stopping on validation F1. For deployment evaluation, we measure inference latency on Jetson Nano, compressed model size, and bandwidth consumption.

4.2 Unimodal Baselines

Table 1. Unimodal backbone comparison on the test set

Model	Precision (%)	Recall (%)	F1 (%)
Image Models			
ResNet-50	**93.79**	**93.81**	**93.75**
DenseNet-121	88.31	88.14	88.14
ResNeXt-50	83.46	83.05	82.82
Text Models			
PhoBERT-base + MLP	**95.56**	**95.59**	**95.59**
CNN (text)	92.46	91.71	91.85
BiLSTM	89.35	88.78	88.88
SVM + TF-IDF	74.04	73.30	73.46

Table 1 reports a systematic comparison of pretrained visual and textual encoders. Among image backbones, ResNet-50 achieves the highest F1-score (93.75%), outperforming DenseNet-121 and ResNeXt-50. Its residual architecture facilitates stable gradient propagation and effective transfer from ImageNet pretraining, yielding superior robustness under real farm conditions. For textual classification, PhoBERT-base significantly surpasses shallow baselines, reaching 95.59% F1. Contextualized transformer embeddings effectively capture subtle semantic variations in symptom descriptions, while traditional models such as SVM and sequence-based BiLSTM struggle with overlapping linguistic cues across disease categories. Based on this comparative study, ResNet-50 and PhoBERT-base are selected as the unimodal backbones for multimodal integration. They provide the best trade-off between representational capacity, transferability, and deployability under edge constraints. Despite not being extremely large models, both demonstrate strong foundation-level generalization when fine-tuned on domain-specific aquaculture data.

4.3 Multimodal Fusion Performance

Late fusion improves performance to 96.45% F1, demonstrating complementary signals between image and text modalities. The proposed cross-modal attention mechanism further increases performance to 98.20% F1, yielding an absolute improvement of +1.75% over late fusion. Although numerically modest, this corresponds to approximately 24 additional correct predictions per 1,000 cases, which is operationally significant in disease-sensitive aquaculture environments.

Table 2. Multimodal fusion performance

Fusion Method	Accuracy (%)	F1 (%)
Late Fusion	96.30	96.45
Cross-Modal Attention	**98.12**	**98.20**

Ambiguity Resolution Analysis Performance gains are most pronounced in three ambiguity scenarios: (1) Visually degraded cases where text clarifies disease identity; (2) Textually ambiguous descriptions where visual evidence disambiguates; (3) Dual-uncertainty cases where attention adaptively reallocates weights across modalities.

To quantitatively validate ambiguity-aware reweighting, we compute the Pearson correlation between modality-specific entropy and corresponding attention weights across the test set in Table 3. For visual modality, we observe a correlation coefficient of $r = -0.62$ ($p < 0.001$), indicating that higher visual uncertainty leads to lower visual attention allocation. Similarly, textual entropy correlates with attention weight at $r = -0.58$ ($p < 0.001$). These statistically significant correlations confirm that the proposed cross-modal attention mechanism systematically downweights high-uncertainty modalities.

Table 3. Correlation between modality entropy and attention weight

Modality	Pearson r	p-value
Image entropy vs. image attention	-0.62	< 0.001
Text entropy vs. text attention	-0.58	< 0.001

4.4 Edge Efficiency and Scalability

Table 4. Edge deployment efficiency

Metric	Value
Model Size (compressed)	15 MB
Inference Latency	< 200 ms
Bandwidth Reduction	70%

After structured pruning and FP16 quantization, the deployed model footprint is reduced to 15 MB while maintaining near-peak accuracy. Inference latency remains below 200 ms on Jetson Nano, enabling real-time, on-device decision support. Bandwidth consumption decreases by approximately 70%, ensuring stable operation under intermittent rural connectivity. These results demonstrate that high-capacity multimodal reasoning can be achieved within strict edge-computing constraints. The framework thus satisfies both accuracy requirements for disease-critical applications and efficiency requirements for scalable IoT deployment, aligning with the objectives of efficient foundation models for large-scale data mining.

4.5 Calibration Analysis.

We evaluate expected calibration error (ECE) and Brier score to assess uncertainty reliability in Table 5. The proposed multimodal attention model achieves the lowest ECE and Brier score, indicating improved probability calibration. This supports our hypothesis that cross-modal interaction reduces predictive entropy and improves uncertainty reliability.

Table 5. Calibration performance comparison

Model	ECE (%)	Brier Score
Image-only	5.63	0.081
Text-only	4.21	0.069
Late Fusion	3.02	0.052
Proposed Model	**1.87**	**0.041**

5 Discussion

Shrimp diseases frequently exhibit overlapping manifestations across modalities. Visually, early-stage WSSV and bacterial infections may share similar discoloration or subtle lesion patterns, especially under turbid water and low illumination. Linguistically, farmer reports such as "reduced feeding" or "lethargic behavior" are semantically non-specific and can correspond to multiple disease categories or environmental stress. These overlaps create intrinsic diagnostic ambiguity when relying on a single modality. Our results demonstrate that multimodal reasoning mitigates this ambiguity by leveraging complementary evidence. The cross-modal attention mechanism adaptively reallocates importance between visual and textual features, effectively reducing predictive uncertainty in cases where one modality is unreliable. This suggests that shrimp disease detection should not be framed as independent unimodal classification, but as structured ambiguity resolution across heterogeneous signals.

Rather than proposing a larger multimodal architecture, this work investigates how pretrained foundation models can be systematically adapted, compressed, and orchestrated within a cyber-physical IoT pipeline. The key insight is that transferability, not scale alone, determines practical foundation model impact in environmental data mining. Our results indicate that pretrained representations retain strong generalization capacity even after structured pruning and quantization, suggesting that efficient specialization is feasible without retraining large-scale models from scratch.

Although validated in shrimp aquaculture, the proposed framework is not domain-specific. The formulation of multimodal ambiguity resolution applies broadly to other settings where heterogeneous signals must be reconciled under uncertainty. Potential extensions include livestock disease detection combining imagery and veterinary notes, crop pathology integrating leaf images with farmer observations, and environmental anomaly mining across sensor streams and textual logs. The approach is particularly relevant for low-resource IoT deployments, where efficiency and robustness are equally critical. Thus, the framework provides a general template for efficient multimodal foundation model deployment in cyber-physical systems.

6 Conclusion

This paper formulates shrimp disease diagnosis as a multimodal ambiguity resolution problem under real-world uncertainty. We proposed an efficient framework that integrates vision and language foundation models through cross-modal attention. Experiments on field data showed an F1-score of 98.2%, outperforming unimodal and late-fusion baselines while remaining suitable for edge deployment through pruning and quantization.

This study has several limitations. The current evaluation is based on a limited operational setting, which may not fully reflect variation across farms, seasons, and environmental conditions. In addition, sensor context was considered at the system level but not yet integrated into the multimodal predictor, and comparisons with larger recent vision-language models were not included. Future work will expand evaluation across more diverse aquaculture settings, incorporate sensor signals directly into the reasoning framework, and benchmark against newer multimodal architectures under edge constraints.

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